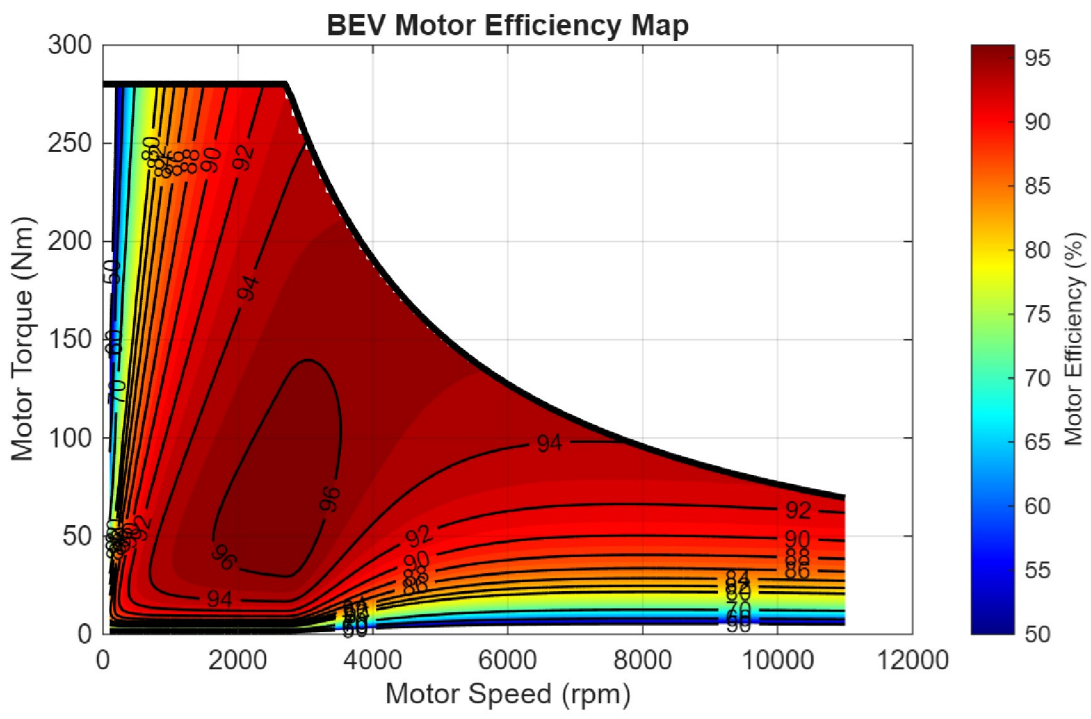
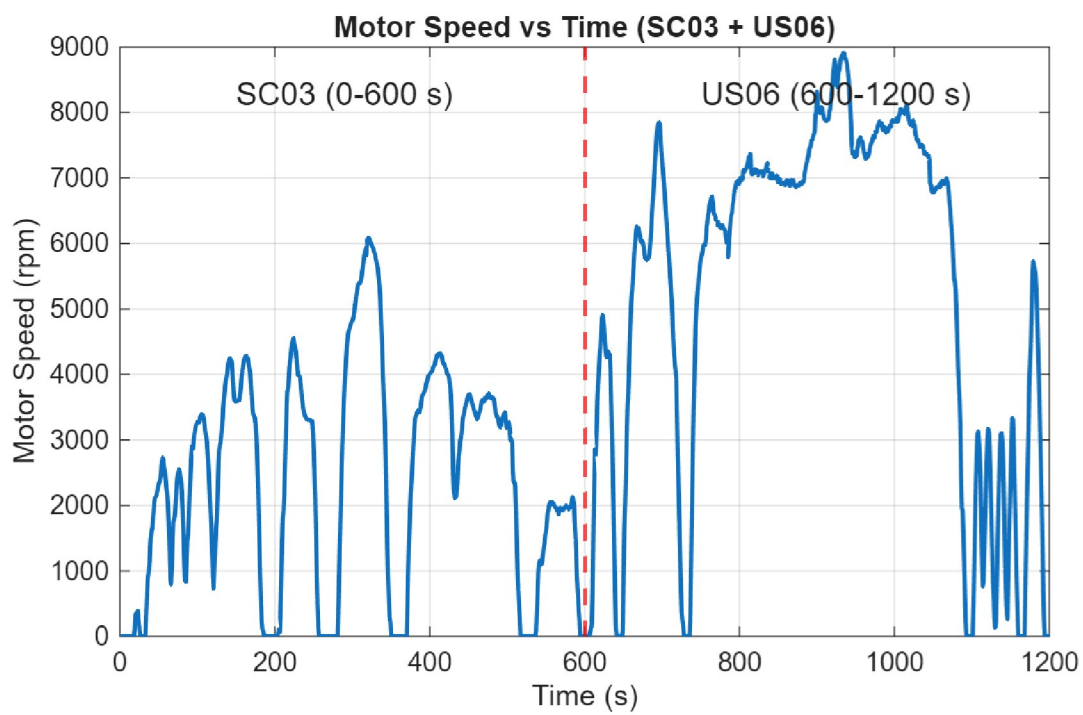
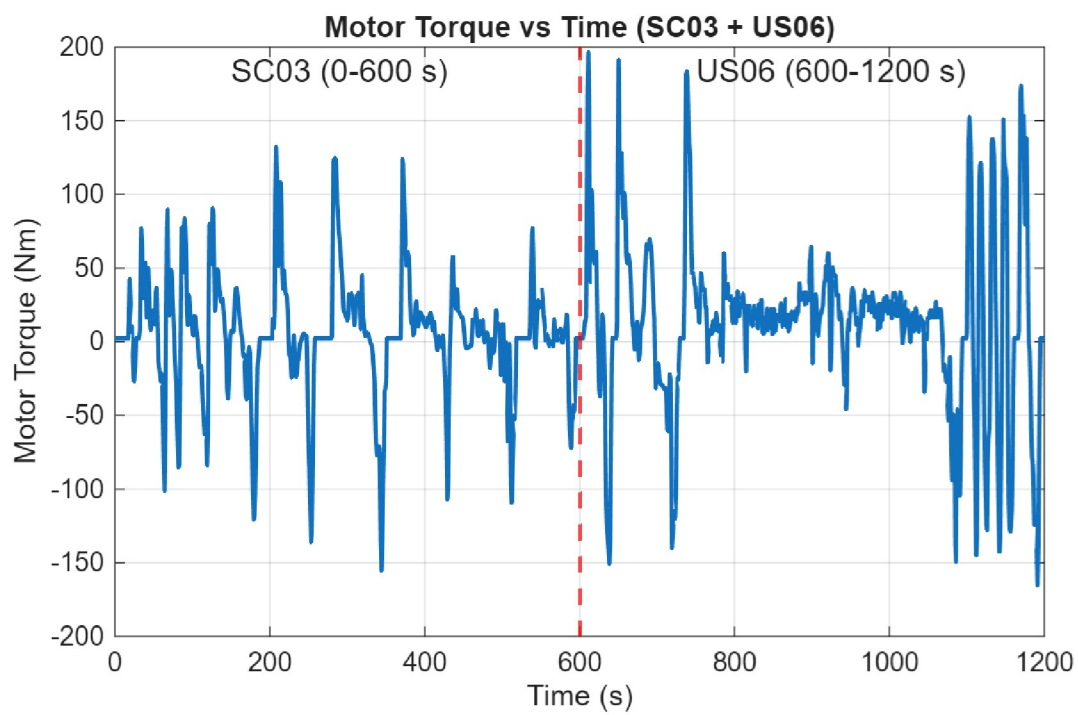
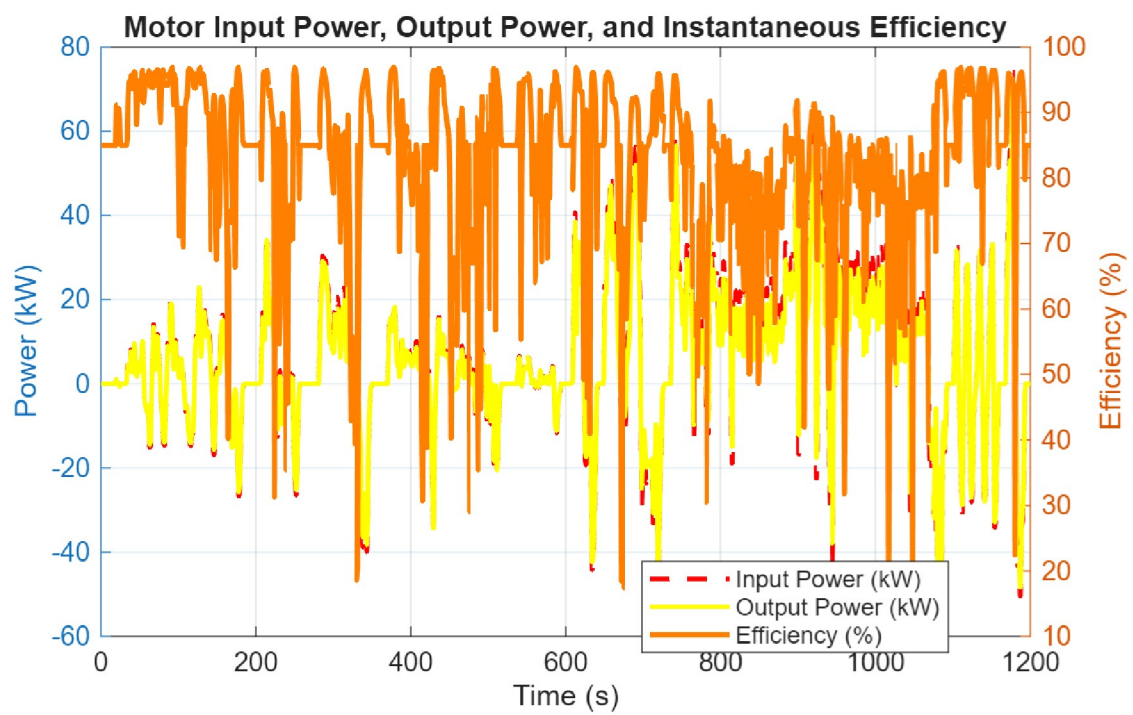
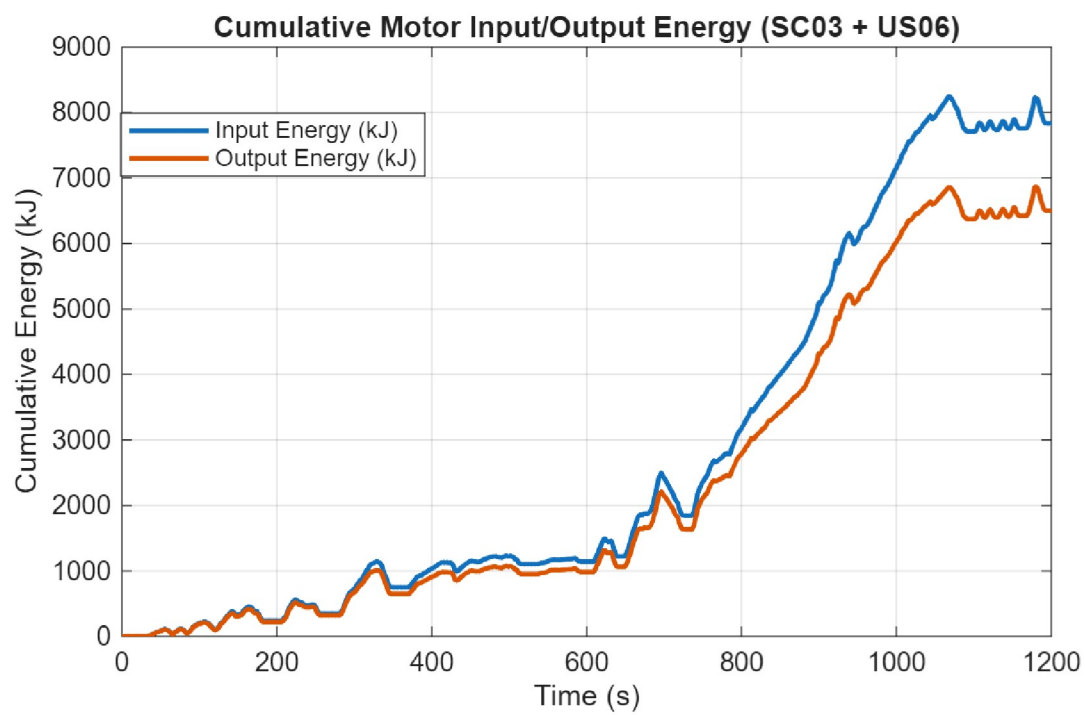












Q1 – Motor Efficiency Map

This figure shows the motor efficiency as a function of speed and torque. The highest efficiency region appears at medium speed and medium torque, while efficiency decreases at very low speed and very high speed. This behavior is physically reasonable because motor losses increase when operating at extreme conditions. The torque limit curve clearly defines the safe operating boundary of the motor. Overall, the efficiency distribution is smooth and realistic, and it matches typical electric motor characteristics.

Q2(i) – Motor Speed vs Time

This figure shows the motor speed during the combined SC03 and US06 drive cycles. In the SC03 section, the motor speed remains relatively low and changes gradually, which is consistent with city driving conditions. In the US06 section, the speed increases significantly and fluctuates more aggressively, representing higher-speed and more dynamic driving. The transition between the two cycles is clear. The overall speed pattern matches the characteristics of the two drive cycles, so the result makes sense.

Q2(ii) – Motor Torque vs Time

This figure shows the motor torque over time. During SC03, the torque changes frequently due to repeated acceleration and deceleration in urban driving. During US06, higher torque peaks appear because of stronger acceleration demands. Negative torque values correspond to braking or regenerative conditions. The maximum torque does not exceed the rated torque of the motor, indicating that a single motor is sufficient for this simplified simulation. The torque trend is consistent with the speed and acceleration behavior, which makes physical sense.

Q3(i) – Input Power, Output Power, and Instantaneous Efficiency

This figure presents the motor input power, output power, and instantaneous efficiency over time. The instantaneous efficiency is not assumed as a constant value. It is calculated by interpolating the efficiency map from Q1 using the motor speed and torque at each time step. Therefore, the efficiency directly depends on the operating point of the motor.

When the vehicle accelerates, both input and output power increase. During braking, power may become negative, representing regenerative operation. The efficiency fluctuates throughout the cycle because the motor operates at different speed–torque combinations. This variation is consistent with the efficiency distribution shown in Q1. The relationship between power and efficiency appears reasonable, so the results make sense.

Q3(ii) – Cumulative Input and Output Energy

This figure shows the accumulated input and output energy over time. Both curves increase as the simulation progresses, and the input energy is always greater than the output energy due to motor losses. During the US06 cycle, the slope becomes steeper, indicating higher energy consumption under high-speed conditions. The difference between the input and output energy represents total motor losses. The smooth and consistent trend indicates that the energy calculation is physically reasonable.

Q3(iii) – Final Energy and Average Efficiency

The final cumulative output energy is **6500.96 kJ**, and the cumulative input energy is **7835.92 kJ**, resulting in an average efficiency of **82.96%**. Since the input energy is greater than the output energy, the difference represents motor losses such as copper loss and mechanical loss. An average efficiency of approximately 83% over a full drive cycle is reasonable for an electric motor operating under varying conditions. Therefore, the results make sense and are physically consistent. However, since this model is simplified and does not include all real-world effects, the results should be considered as an approximation rather than exact vehicle data.